

Final Web Development Project Requirements

Your final project is to build a complete, working web application. It should feel like a real product, not a collection of disconnected pages. The goal is to show that you can plan, build, revise, and explain a web app that uses data, user interaction, and at least one meaningful AI-powered feature.

Objective Requirements

1. GitHub Checkpoints

Before you start building, plan your project around at least **10 checkpoints or milestones**. These checkpoints should help you build in a smart order and give you safe rollback points if something breaks.

Each checkpoint must:

1. Represent a major working step in the project.
2. Be tested by the human developer before moving on.
3. Leave the rest of the app working as expected.
4. Be labeled using this format: **#01-Descriptive-Title**.
5. Be recorded on a Checkpoints page in your app.

Your **Checkpoints page** must include one entry for each checkpoint with:

1. Checkpoint name
2. Date completed
3. Link to the GitHub commit
4. Brief reflection on what you built, fixed, or learned

A checkpoint should not be a tiny change like changing a color, fixing one typo, or moving one button. It should show that a complete feature or important part of the app is working.

Examples of good checkpoints:

- **#01-Project-Setup:** Files organized and app running
- **#02-Navigation-Routing:** Main layout, navigation, and routes working
- **#03-Home-Page:** Home page shows the app concept clearly
- **#04-Registration:** Registration flow working
- **#05-User-Profile:** Profile page displays saved user data

- **#06-Extended-Profile:** Extended profile form saves required profile fields
 - **#07-Create-Data:** Users can create main app data
 - **#08-View-Data:** User-created data displays correctly
 - **#09-Edit-Data:** Users can edit saved data
 - **#10-AI-Feature:** AI feature uses user input or app data
 - **#11-Admin-Dashboard:** Admin dashboard displays app data
 - **#12-Admin-Update:** Admin can update app data
 - **#13-Final-Polish:** Styling, bug fixes, and usability improvements complete
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2. Minimum Pages or Views

Your app must include at least **8 pages or views**.

A page or view is a distinct screen, route, or major section of the app that has its own purpose.

Examples:

- Home page
- Profile page
- Create or interact with an item page
- Admin dashboard
- Admin edit page

A small pop-up, button, or repeated card does not count as a full page or view unless it represents a major app section.

3. Data Files [JSON]

Your project must include at least **3 JSON data files**. These files should act as the starting data for your app.

Your JSON files must include:

1. **User data**
 - Stores registration, profile, or extended profile information.
 - Example file: `users.json`
2. **Main app data**
 - Stores the main thing your app is built around.
 - Examples: recipes, workouts, study tasks, products, events, reviews, goals, lessons, or projects.
 - Example file: `items.json`, `recipes.json`, `workouts.json`, or `tasks.json`

3. Admin or app settings data

- Stores information that an admin can manage or update.
- Examples: categories, featured items, approval status, difficulty levels, announcements, tags, or app settings.
- Example file: `categories.json`, `settings.json`, or `announcements.json`

Each JSON file must contain:

1. At least **5 records**
2. At least **4 fields per record**
3. A unique `id` field for each record
4. Clear field names that match the purpose of the app
5. Valid JSON formatting

At least one JSON file must connect to another JSON file using an ID.

Example:

- A user has an `id` of `user-001`
- A saved recipe has a `userId` of `user-001`
- This shows which user created or saved the recipe

Your data should not be random filler. It should support your pages, user actions, admin dashboard, and AI-powered feature.

4. AI-Powered Feature

Your project must include at least **1 AI-powered feature**.

This feature must do something useful inside your app that is specifically related to you app. A generic chatbot by itself does **not** count.

A strong AI feature uses information from the app or the user to produce a helpful result.

5. User Actions Must Create and Edit Data

Users must be able to **create data** and **edit data** in your app beyond the user profile page.

This means your app should include forms or controls that allow users to add information and later change it, or your app tracks user interactions on your webapp for use in other features.

6. Registration and Profile

Your app must include a **registration process** and a **user profile**.

The registration process should collect basic account or user information. The profile should display information about the user after they register.

Minimum expectations:

- A registration page or form
- A profile page or view
- Profile information that appears after registration
- A way for the user to update at least part of their profile

Depending on the tools used in class, this may use real authentication, simulated authentication, localStorage, Firebase, Supabase, or another approved method.

7. Extended Profile

Your app must include an **extended profile** with at least **9 profile data items**. Choose fields that fit your app. Do not add random profile fields just to reach the number. The extended profile should connect to the purpose of the app, especially if your AI feature uses this information.

8. Admin Dashboard

Your app must include an **admin dashboard**. The admin dashboard is a separate view for managing or reviewing app data. Regular users must not have access to this view or functionality. The admin dashboard should look and function differently from a normal user page.

9. Admin Update Data

The admin must be able to **update app data**. This means the admin dashboard cannot only display information. It must allow the admin to make a change. The update must be visible by regular users somewhere in the app after it is made.

Final Project Success Criteria

A successful project should clearly demonstrate that you can:

- Build multiple connected pages or views
- Use GitHub to track progress over time
- Create forms that collect useful data
- Save and display user-created information
- Allow users to edit information
- Build a registration and profile experience
- Use profile data in a meaningful way
- Add an AI-powered feature connected to the app's purpose
- Create an admin dashboard
- Allow an admin to update data
- Explain how the app works

Important Reminder

Your project should solve a clear problem or serve a clear purpose. Do not build random pages just to meet the number requirements. Every page, form, data field, and AI feature should support the main idea of your app.